The Norm of a Multiplication Operator on L^2

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Abstract

We prove in detail that if $(X, \mathcal{B}, \mu)$ is a σ-finite measure space and $\varphi \in L^\infty(\mu)$, then the multiplication operator $M_\varphi : L^2(X) \to L^2(X)$ defined by $M_\varphi(f) = \varphi f$ is linear and

$$\|M_\varphi\| = \|\varphi\|_\infty,$$

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1 Preliminaries and notation

Let $(X, \mathcal{B}, \mu)$ be a measure space. We denote by $L^2(X) = L^2(X, \mathcal{B}, \mu)$ the usual Hilbert space of (equivalence classes of) measurable functions $f : X \to \mathbb{C}$ with $\int_X |f|^2 \, d\mu < \infty$. Norm on L^2 is $\|f\|_2 = (\int_X |f|^2 \, d\mu)^{1/2}$.

Definition 1.1. *For a measurable function $\varphi : X \to \mathbb{C}$*

$$\|\varphi\|_\infty := \inf\{C \geq 0 : \mu(\{x \in X : |\varphi(x)| > C\}) = 0\}.$$

When $\|\varphi\|_\infty < \infty$, we write $\varphi \in L^\infty(\mu)$.

Definition 1.2 (Multiplication operator)**.** *For $\varphi \in L^\infty(\mu)$ define*

$$M_\varphi : L^2(X) \to L^2(X), \qquad (M_\varphi f)(x) = \varphi(x) f(x).$$

M_φ is linear and well-defined

$$\|M_\varphi f\|_2^2 = \|f\varphi\|_2^2 \leq \int_E |f\varphi|^2 d\mu \leq \|\varphi\|_\infty^2 \int_E |f|^2 d\mu = \|\varphi\|_\infty^2 \|f\|_2^2 < \infty$$

so $M_\varphi f \in L^2(X)$

2 M_φ is Bounded Linear operator

Proposition 2.1. *Let $(X, \mathcal{B}, \mu)$ be a measure space (sigma-finite) and let $\varphi \in L^\infty(X, \mu)$. Define the multiplication operator $M_\varphi : L^2(X, \mu) \to L^2(X, \mu)$ by*

$$M_\varphi(f) := \varphi f, \qquad \forall f \in L^2(X, \mu).$$

Then M_φ is a bounded linear operator on $L^2(X, \mu)$.

Proof. for $f, g \in L^2(X, \mu)$ and scalars a, b,

$$M_\varphi(af + bg) = \varphi(af + bg) = a\varphi f + b\varphi g = aM_\varphi(f) + bM_\varphi(g).$$

Next compute the L^2-norm of $M_\varphi(f)$. Since $\varphi \in L^\infty$, we have $|\varphi(x)| \leq \|\varphi\|_\infty$ for almost every x, hence

$$\begin{aligned} \|M_\varphi(f)\|_2^2 &= \int_X |\varphi f|^2 \, d\mu = \int_X |\varphi|^2 \, |f|^2 \, d\mu \\ &\leq \|\varphi\|_\infty^2 \int_X |f|^2 \, d\mu = \|\varphi\|_\infty^2 \|f\|_2^2. \end{aligned}$$

Taking square roots gives the operator bound

$$\|M_\varphi(f)\|_2 \leq \|\varphi\|_\infty \|f\|_2 \qquad \text{for all } f \in L^2(X, \mu).$$

Therefore M_φ is a bounded linear operator on $L^2(X, \mu)$. □

3 Upper bound: $\|M_\varphi\| \leq \|\varphi\|_\infty$

Proposition 3.1. *If $\varphi \in L^\infty(\mu)$, then for every $f \in L^2(X)$*

$$\|M_\varphi f\|_2 \leq \|\varphi\|_\infty \|f\|_2,$$

hence $\|M_\varphi\| \leq \|\varphi\|_\infty$.

Proof. For $f \in L^2$ we have

$$\|M_\varphi f\|_2^2 = \int_X |\varphi(x) f(x)|^2 \, d\mu(x) \leq \|\varphi\|_\infty^2 \int_X |f(x)|^2 \, d\mu(x) = \|\varphi\|_\infty^2 \|f\|_2^2.$$

Taking square roots gives the desired inequality. Taking the supremum over unit vectors $\|f\|_2 = 1$ yields $\|M_\varphi\| \leq \|\varphi\|_\infty$. □

4 Lower bound: strategy

To show equality, we need $\|M_\varphi\| \geq \|\varphi\|_\infty$. The standard strategy is:

- For any $\alpha < \|\varphi\|_\infty$, the set $E_\alpha := \{x : |\varphi(x)| \geq \alpha\}$ has positive measure (by definition of essential supremum).
- Construct a unit vector $f_\alpha \in L^2$ supported inside E_α, typically $f_\alpha = \chi_{E_\alpha} / \sqrt{\mu(E_\alpha)}$.
- Compute $\|M_\varphi f_\alpha\|_2$ and show it is at least α.
- Since the inequality holds for every $\alpha < \|\varphi\|_\infty$, conclude $\|M_\varphi\| \geq \|\varphi\|_\infty$.

5 Equality of the operator norm of a multiplication operator for X finite-measure case

Assume $\mu(X) < \infty$.

Proposition 5.1. *If $\mu(X) < \infty$ and $\varphi \in L^\infty(\mu)$ then $\|M_\varphi\| = \|\varphi\|_\infty$.*

Proof. We already have $\|M_\varphi\| \leq \|\varphi\|_\infty$ To prove the reverse inequality.If $\|\varphi\|_\infty = 0$ the result is trivial, so assume $\|\varphi\|_\infty > 0$. let α be any number with $0 \leq \alpha < \|\varphi\|_\infty$. By the definition of essential supremum, the set

$$E_\alpha := \{x \in X : \ |\varphi(x)| \geq \alpha\}$$

has positive measure: $\mu(E_\alpha) > 0$. Since $\mu(X) < \infty$ we also have $\mu(E) < \infty$ (By monotonicity of measure).

Define

$$f_\alpha(x) := \frac{\chi_{E_\alpha}(x)}{\sqrt{\mu(E_\alpha)}}.$$

Then Then

$$\|f_\alpha\|_2^2 = \int_X |f_\alpha|^2 \, d\mu = \int_X \frac{\chi_{E_\alpha}^2}{\mu(E_\alpha)} \, d\mu = \frac{1}{\mu(E_\alpha)} \int_X \chi_{E_\alpha} \, d\mu = \frac{\mu(E_\alpha)}{\mu(E_\alpha)} = 1,$$

so $\|f_\alpha\|_2 = 1$.

$f_\alpha \in L^2(X)$ and $\|f_\alpha\|_2 = 1$.
Compute

$$\begin{aligned}\|M_\varphi f_\alpha\|_2^2 &= \int_X |\varphi f_\alpha|^2 \, d\mu = \int_X |\varphi|^2 \, |f_\alpha|^2 \, d\mu = \int_X |\varphi|^2 \frac{\chi_{E_\alpha}}{\mu(E_\alpha)} \, d\mu = \int_{E_\alpha} |\varphi|^2 \frac{1}{\mu(E_\alpha)} \, d\mu \\ &\geq \int_{E_\alpha} \frac{\alpha^2}{\mu(E_\alpha)} \, d\mu = \frac{\alpha^2 \mu(E_\alpha)}{\mu(E_\alpha)} = \alpha^2.\end{aligned}$$

Since $|\varphi(x)| \geq \alpha$ for a.e. $x \in E_\alpha$, the integrand satisfies $|\varphi(x)|^2 \geq \alpha^2$ a.e. on E_α. Therefore

$$\|M_\varphi f_\alpha\|_2^2 \geq \frac{1}{\mu(E_\alpha)} \int_{E_\alpha} \alpha^2 \, d\mu(x) = \alpha^2,$$

so $\|M_\varphi f_\alpha\|_2 \geq \alpha$. Since $\|f\|_2 = 1$ by the definition of operator norm (supremum over unit vectors),

$$\|M_\varphi\| \geq \alpha.$$

Because this holds for every $0 \leq \alpha < \|\varphi\|_\infty$, by letting $\alpha \uparrow \|\varphi\|_\infty$ we conclude $\|M_\varphi\| \geq \|\varphi\|_\infty$. Combined with the reverse bound $\|M_\varphi\| \leq \|\varphi\|_\infty$, we conclude $\|M_\varphi\| = \|\varphi\|_\infty$. □

6 Equality of the operator norm of a multiplication operator for sigma-finite case

Proposition 6.1. *If $(X, \mathcal{B}, \mu)$ is sigma-finite and $\varphi \in L^\infty(X, \mu)$. Define the multiplication operator $M_\varphi : L^2(X, \mu) \to L^2(X, \mu)$ by $M_\varphi(f) := \varphi f$.*
then

$$\|M_\varphi\| = \|\varphi\|_\infty.$$

Proof. We already know the general bound $\|M_\varphi\| \le \|\varphi\|_\infty$. The goal of the proposition below is to prove the reverse inequality.
If $\|\varphi\|_\infty = 0$ the result is trivial, so assume $\|\varphi\|_\infty > 0$. Fix any number $0 \le \alpha < \|\varphi\|_\infty$. By the definition of the essential supremum, the measurable set

$$E_\alpha := \{x \in X : \ |\varphi(x)| > \alpha\}$$

has positive measure: $\mu(E_\alpha) > 0$.

Because $(X, \mathcal{B}, \mu)$ is sigma-finite there exists a sequence of measurable sets $(X_n)_{n\ge1}$ with $\mu(X_n) < \infty$ and $X = \bigcup_{n=1}^{\infty} X_n$. Since $E_\alpha = \bigcup_{n=1}^{\infty}(E_\alpha \cap X_n)$ and $\mu(E_\alpha) > 0$, There must exist some index n for which

$$\mu(E_\alpha \cap X_n) > 0.$$

Set

$$E := E_\alpha \cap X_n.$$

Then $E \subset E_\alpha$ is measurable and satisfies $0 < \mu(E) < \mu(X_n) < \infty$. (This is the finite-measure subset of E_α whose existence is guaranteed by sigma-finiteness.)

Define

$$f_\alpha(x) := \frac{\chi_E(x)}{\sqrt{\mu(E)}},$$

where χ_E is the charachteristic function of E. A direct calculation shows

$$\|f_\alpha\|_2^2 = \int_X |f_\alpha|^2 \, d\mu = \int_X \frac{\chi_E}{\mu(E)} \, d\mu = \frac{1}{\mu(E)} \int_X \chi_{E_\alpha} \, d\mu = \frac{\mu(E)}{\mu(E)} = 1,$$

so $\|f_\alpha\|_2 = 1$.

Compute the norm of the image under M_φ:

$$\begin{aligned}
\|M_\varphi f_\alpha\|_2^2 &= \int_X |M_\varphi f_\alpha(x)|^2 \, d\mu(x) && \text{(definition of the } L^2\text{norm)} \\
&= \int_X |\varphi(x)\, f_\alpha(x)|^2 \, d\mu(x) && \text{(definition of } M_\varphi) \\
&= \int_X |\varphi(x)|^2 \, |f_\alpha(x)|^2 \, d\mu(x) && \text{(absolute value distributes over products)} \\
&= \int_X |\varphi(x)|^2 \left| \frac{\chi_E(x)}{\sqrt{\mu(E)}} \right|^2 d\mu(x) && \text{(definition of } f_\alpha) \\
&= \int_X |\varphi(x)|^2 \frac{\chi_E(x)}{\mu(E)} \, d\mu(x) && \text{(since } \chi_E^2 = \chi_E) \\
&= \int_E |\varphi(x)|^2 \, \frac{1}{\mu(E)} \, d\mu(x) && \text{(restriction of the integral to } E) \\
&\ge \int_E \frac{\alpha^2}{\mu(E)} \, d\mu(x) && \text{(because } |\varphi(x)| \ge \alpha \text{ on } E \subset E_\alpha) \\
&= \frac{\alpha^2}{\mu(E)} \int_E 1 \, d\mu(x) && \text{(pull out constants)} \\
&= \frac{\alpha^2}{\mu(E)} \, \mu(E) && \text{(measure of } E) \\
&= \alpha^2.
\end{aligned}$$

because $E \subset E_\alpha$ and thus $|\varphi(x)| > \alpha$ for a.e. $x \in E$. Hence $\|M_\varphi f_\alpha\|_2 \geq \alpha$. Since $\|f_\alpha\|_2 = 1$, the definition of the operator norm gives $\|M_\varphi\| \geq \alpha$.

Because the choice of α was any number with $0 \leq \alpha < \|\varphi\|_\infty$,by letting $\alpha \uparrow \|\varphi\|_\infty$ we obtain $\|M_\varphi\| \geq \|\varphi\|_\infty$. Combining this with the already-known bound $\|M_\varphi\| \leq \|\varphi\|_\infty$ yields the desired equality

$$\|M_\varphi\| = \|\varphi\|_\infty.$$

□